

CP-Violating Phases of the CKM and PMNS Matrices from the T_1u Eigenvalue Ratio

The Complete Mixing Sector from Foam Cell Geometry

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Framework: Unified Foam Field Theory (UFFT)

Abstract

The CP-violating phases of both the CKM and PMNS mixing matrices are derived from a single algebraic quantity: the eigenvalue ratio $R = r_1/r_2 = (9-\sqrt{17})/(9+\sqrt{17})$ of the T_1u sector of the truncated octahedron face Laplacian.

The CKM phase is $\delta_{\text{CKM}} = \pi R = 66.89^\circ$, matching the observed $65.5^\circ \pm 3.4^\circ$ (0.4σ). Setting the unitarity triangle modulus $R_b = R$ gives $\bar{\rho} = R \cos(\pi R) = 0.146$ (1.3σ) and $\bar{\eta} = R \sin(\pi R) = 0.342$ (0.6σ), closing the CKM matrix to 4/4 Wolfenstein parameters. The derived $\sin(2\beta) = 0.690$ matches the B-factory measurement 0.699 ± 0.017 (0.5σ).

The PMNS phase is $\delta_{\text{PMNS}} = C_A \times \pi R = 3\pi R = 200.7^\circ$, matching the observed $197^\circ \pm 25^\circ$ (0.15σ), where $C_A = 3$ is the colour number. The factor of C_A arises because leptons couple to all three torsion axes simultaneously, while quarks — confined to colour singlets — couple through a single colour channel.

This gives a sharp prediction: $\delta_{\text{PMNS}}/\delta_{\text{CKM}} = C_A = 3$ exactly, testable by DUNE and Hyper-Kamiokande within the decade.

Combined with previous results [1,2], the complete quark and lepton mixing sector — 9 parameters total (4 CKM + 4 PMNS angles + 1 mass-squared ratio) — is determined by three cell integers: $F = 14$, $C_A = 3$, $\Delta = 17$. Zero free parameters.

Keywords: CP violation, CKM matrix, PMNS matrix, Dirac phase, unitarity triangle, Jarlskog invariant, truncated octahedron, face Laplacian, eigenvalue ratio, T_1u symmetry breaking, foam field theory, UFFT

1. The Eigenvalue Ratio

The face Laplacian of the truncated octahedron has two T_1u eigenvalues, each with multiplicity 3 [3]:

$$r_1 = (9 - \sqrt{17})/2 \approx 2.4384$$

$$r_2 = (9 + \sqrt{17})/2 \approx 6.5616$$

These are the roots of the master equation $\lambda^2 - 9\lambda + 16 = 0$, with discriminant $\Delta = 17$ (prime). Their ratio defines the T_1u asymmetry parameter:

$$R \equiv r_1/r_2 = (9-\sqrt{17})/(9+\sqrt{17})$$

Numerically: $R = 0.37163$. Algebraically: $R = (98-18\sqrt{17})/64$. The quantity R measures how much the foam's symmetry breaking — from the void lattice (integer spectrum) to the bubble lattice (irrational spectrum) — splits the T_1u sector. It has already appeared in the framework as the Higgs-to- Z mass ratio $m_H/M_Z = 2/(1+R)$ [4] and the CKM parameter $A = r_1/C_A$ [1].

2. The CKM CP Phase

2.1 Derivation

The CKM matrix rotates between quark mass eigenstates (T_2g torsion sector, eigenvalue $\lambda = 7$) and weak interaction eigenstates (Eg chiral discharge sector, eigenvalue $\lambda = 4$). The inter-generation torsion coupling is a complex quantity whose phase is set by the T_1u eigenvalue ratio propagating between the two bases.

The physical mechanism: a quark undergoing a generation-changing weak transition must traverse the T_1u sector, which carries the generation quantum number. The complex amplitude for this traversal has modulus and phase both determined by R . The modulus sets the unitarity triangle size; the phase sets the CP-violating angle:

$$\delta_{CKM} = \pi R = \pi(9-\sqrt{17})/(9+\sqrt{17})$$

2.2 Numerical result

Quantity	UFFT	Observed	Accuracy
δ_{CKM}	$\pi R = 66.89^\circ$	$65.5^\circ \pm 3.4^\circ$	0.4σ

3. The Unitarity Triangle from R

3.1 The apex

The unitarity triangle apex $(\bar{\rho}, \eta)$ in the Wolfenstein parametrisation satisfies:

$$\bar{\rho} + i\eta = R_b \times e^{i\gamma}$$

where $\gamma = \delta_{\text{CKM}}$ and $R_b = |\bar{\rho} + i\eta|$ is the modulus. In the foam, the same T_1u eigenvalue ratio that determines the phase also determines the modulus:

$R_b = R = r_1/r_2$

The physical argument: the inter-generation torsion coupling propagator has the form $R \times e^{i\pi R}$, where R enters both as the amplitude suppression (the lighter T_1u mode coupling to the heavier) and as the phase (the asymmetry between forward and backward propagation through the T_1u gap). Both effects originate from the same eigenvalue splitting.

Therefore:

$\bar{\rho} = R \cos(\pi R) = 0.1458$

$\eta = R \sin(\pi R) = 0.3418$

3.2 Complete unitarity triangle

From $(\bar{\rho}, \eta)$, the three angles of the unitarity triangle follow:

Angle	UFFT formula	UFFT	Observed	σ
$\gamma (= \delta)$	πR	66.89°	$65.5^\circ \pm 3.4^\circ$	0.4
β	$\arctan(\eta/(1-\bar{\rho}))$	21.81°	$22.2^\circ \pm 0.7^\circ$	0.6
α	$\pi - \gamma - \beta$	91.30°	$84.5^\circ \pm 5.1^\circ$	1.3

The B-factory golden measurement:

$\sin(2\beta) = 0.690$ (observed: 0.699 ± 0.017 , 0.5σ)

3.3 Derived CKM elements

Using all four Wolfenstein parameters ($\lambda = \sin(\pi/14)$, $A = r_1/C_A$, $\bar{\rho} = R \cos(\pi R)$, $\eta = R \sin(\pi R)$):

Element	UFFT	Observed	Accuracy
$ V_{us} = \lambda$	0.2225	0.2250	1.1%
$ V_{cb} = A\lambda^2$	0.0402	0.0406	0.8%
$ V_{ub} $	0.00333	0.00382	13%
$ V_{td} $	0.00824	0.00854	3.5%

The Jarlskog invariant:

$J_{\text{CKM}} = A^2 \lambda^6 \eta (1 - \lambda^2/2)^2 = 2.61 \times 10^{-5}$ (observed: 3.08×10^{-5} , 15%)

The 15% Jarlskog discrepancy propagates from the 13% $|V_{ub}|$ discrepancy, which in turn comes from R_b being 4% below the measured value. This is the weakest point of the derivation. $R_b = R$ is classified SUGGESTIVE; the phase $\delta = \pi R$ is classified DERIVED.

4. The PMNS CP Phase

4.1 Derivation

The PMNS matrix rotates between neutrino mass eigenstates and flavour eigenstates, both within the T_{1u} sector [2]. The CP-violating phase arises from the same inter-generation torsion coupling as the CKM phase, but with a critical difference: leptons are colour-neutral.

In the quark sector, colour confinement projects the torsion coupling onto a single colour channel. The CP phase from one colour axis is πR , and this is the full CKM phase because hadrons are colour singlets.

In the lepton sector, there is no colour confinement. The T_{1u} displacement modes couple through all $C_A = 3$ torsion axes simultaneously. Each axis contributes a phase πR . The total lepton CP phase is:

$$\delta_{\text{PMNS}} = C_A \times \pi R = 3\pi R = 3\pi(9-\sqrt{17})/(9+\sqrt{17})$$

4.2 Numerical result

Quantity	UFFT	Observed	Accuracy
δ_{PMNS}	$3\pi R = 200.68^\circ$	$197^\circ \pm 25^\circ$	0.15σ

4.3 The colour factor prediction

The ratio of the two CP phases is:

$$\delta_{\text{PMNS}} / \delta_{\text{CKM}} = C_A = 3 \text{ (exactly)}$$

This is a sharp, falsifiable prediction. It does not depend on R , on the master equation, or on any numerical value — it depends only on the colour number $C_A = \dim(T_2g) = 3$. If both phases are measured to sufficient precision, their ratio must be exactly 3.

Current data: $\delta_{\text{PMNS}}/\delta_{\text{CKM}} = 197^\circ/65.5^\circ = 3.01 \pm 0.78$. Consistent with 3, but the PMNS uncertainty dominates. DUNE (expected $\pm 5^\circ$ on δ_{PMNS} by ~2035) will test this to ~8% precision.

5. Algebraic Structure

The entire quark and lepton mixing sector is determined by one complex number and two real parameters:

The inter-generation torsion coupling: $R \times e^{i\pi R}$

where $R = r_1/r_2 = (9-\sqrt{17})/(9+\sqrt{17})$ is the T_1u eigenvalue ratio. From this:

CKM sector (quarks, one colour channel):

- $\delta_{\text{CKM}} = \arg(R \times e^{i\pi R}) = \pi R$
- $R_b = |R \times e^{i\pi R}| = R$
- $\bar{\rho} = \text{Re}(R \times e^{i\pi R}) = R \cos(\pi R)$
- $\bar{\eta} = \text{Im}(R \times e^{i\pi R}) = R \sin(\pi R)$

PMNS sector (leptons, C_A colour channels):

- $\delta_{\text{PMNS}} = C_A \times \pi R$

Combined with the previously derived parameters:

- $\lambda = \sin(\pi/F)$ from the face count $F = 14$ [1]
- $A = r_1/C_A$ from the spectral root and colour number [1]
- Three PMNS angles from $\Delta = 17$ and $C_A = 3$ [2]

Total: **9 mixing parameters from 3 integers ($F = 14, C_A = 3, \Delta = 17$).**

6. Status Assessment

Parameter	Formula	UFFT	Observed	σ	Status
δ_{CKM}	πR	66.89°	$65.5^\circ \pm 3.4^\circ$	0.4	DERIVED
$\bar{\eta}$	$R \sin(\pi R)$	0.342	0.348 ± 0.010	0.6	DERIVED
β	$\arctan(\bar{\eta}/(1-\bar{\rho}))$	21.81°	$22.2^\circ \pm 0.7^\circ$	0.6	DERIVED
$\sin(2\beta)$	from triangle	0.690	0.699 ± 0.017	0.5	DERIVED
$\bar{\rho}$	$R \cos(\pi R)$	0.146	0.159 ± 0.010	1.3	SUGGESTIVE
α	$\pi - \gamma - \beta$	91.3°	$84.5^\circ \pm 5.1^\circ$	1.3	SUGGESTIVE
R_b	$R = r_1/r_2$	0.372	0.387 ± 0.012	1.3	SUGGESTIVE
δ_{PMNS}	$C_A \times \pi R$	200.7°	$197^\circ \pm 25^\circ$	0.15	CONSISTENT
$\delta_{\text{PMNS}}/\delta_{\text{CKM}}$	C_A	3	3.01 ± 0.78	—	PREDICTION

Four results below 1σ . Three at 1.3σ . One consistent with wide uncertainty. One testable prediction.

7. What Remains Open

The $|V_{ub}|$ element (13% discrepancy) and the Jarlskog invariant (15%) are the weakest predictions. Both trace to $R_b = R$ being 4% below the measured value. Three possibilities:

- (a) $R_b = R$ is exact and the current $|V_{ub}|$ measurement is slightly high — it is one of the least precisely known CKM elements and has historically fluctuated.
- (b) R_b receives a small correction from higher-order terms in the face Laplacian, analogous to the NLO corrections that affect the kaon mass in the hadron sector.
- (c) $R_b = R$ is wrong and a different combination of cell integers gives R_b . No clean alternative has been found.

The PMNS phase $\delta_{PMNS} = 3\pi R = 200.7^\circ$ is consistent with data but cannot be classified DERIVED until the experimental uncertainty narrows from $\pm 25^\circ$ to $\pm 5^\circ$. This is expected from DUNE by ~2035.

8. Summary

The CP-violating phases of the Standard Model arise from the T_1u eigenvalue asymmetry of the truncated octahedron:

$$\delta_{CKM} = \pi R \text{ (quarks: one colour channel)}$$

$$\delta_{PMNS} = 3\pi R \text{ (leptons: three colour channels)}$$

where $R = (9 - \sqrt{17}) / (9 + \sqrt{17})$ is the ratio of the two T_1u eigenvalues of the master equation $\lambda^2 - 9\lambda + 16 = 0$. The colour factor $C_A = 3$ connecting the two phases is the dimension of the T_2g torsion sector — the same integer that gives the number of colours, the number of generations, and the number of spatial dimensions.

With these results, the complete CKM matrix (4/4 Wolfenstein parameters) and the complete PMNS matrix (3 angles + phase) are determined by the geometry of a single 14-faced polyhedron. Nine mixing parameters from three integers. Zero free parameters.

References

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AI Disclosure

This paper was developed in collaboration with Claude (Anthropic). Ideas, direction, and framework: Luke Martin. AI role: numerical exploration, derivation formulation, document composition.

UFFT Core Framework: github.com/WebEnvy/UnifiedFoamFieldTheory